

Chapter 166 — Management of Castration-Resistant Prostate Cancer

IF YOU READ NOTHING ELSE

- **Castrate testosterone <50 ng/dL; continue LHRH agent indefinitely** in CRPC.
- **nmCRPC treat if PSADT ≤10 mo** → apalutamide, enzalutamide or darolutamide.
- **Abiraterone 1000 mg fasting + prednisone**; enzalutamide needs no steroid.
- **Docetaxel 75 mg/m² 3-weekly** = first-line chemotherapy (TAX 327).
- **Cabazitaxel 20 mg/m² post-docetaxel**; beats second ARPI (CARD).
- **ARPI** → ARPI works in only 15–30%: switch mechanism.
- **Radium-223**: symptomatic bone-only; **never with abiraterone**.
- **¹⁷⁷Lu-PSMA-617** after ARPI + taxane; PSMA PET-positive (VISION).
- **PARPI benefit greatest in BRCA2**; rucaparib not for ATM.
- **Cord compression** → IV dexamethasone → MRI → surgery + RT.

CLINICAL CONSIDERATIONS

- **Framework**: PCWG3 clinical states model, not TNM.
- **First sign of progression**: usually rising PSA — but not after early ARPI (ARCHES).
- **Antiandrogen withdrawal**: stop first-generation antiandrogen, watch PSA **4–8 weeks**.
- **Check testosterone**: suspected non-compliance; antiandrogen, oestrogen or 5α-reductase-inhibitor monotherapy.
- **PSADT <3 mo** → aggressive course → treat aggressively.
- **Bone pain or visceral-predominant** → upfront chemotherapy.
- **Somatic NGS**: tumour tissue (usually primary); liquid biopsy false-positives from CHIP.
- **Germline testing**: all metastatic; BRCA/Lynch/HOXB13 family history; Ashkenazi.

NONMETASTATIC CRPC

- **Predictors**: PSA kinetics; baseline **PSA >10 ng/mL** + PSA velocity.
- **Natural history**: ~1/3 bone metastasis at 2 yr; median ~30 mo.
- **Approved**: apalutamide (first), enzalutamide, darolutamide. **Abiraterone not approved**.

METASTATIC CRPC

- **Visceral <20%**; nodal 30–40%; bone overwhelmingly commonest.
- **Preferred end point**: PFS (radiographic), not response rate.
- **Commonest haematological problem**: anaemia; avoid ESAs (↑ mortality).
- **Thrombocytosis**: clotting rare; no treatment.
- **Obstructed kidneys**: stent/nephrostomy before renally cleared chemotherapy.

EVALUATION OF TREATMENT EFFICACY

- **PCWG3 bone scan**: new baseline at **12–16 wk**; progression = ≥2 new lesions.
- **Flare**: PSA (radium-223, Lu-PSMA), bone scan, PSMA PET (~13%, resolves by 3 mo).
- **Soft tissue**: RECIST.
- **No PSA fall expected** with sipuleucel-T or radium-223.

NEUROENDOCRINE / AGGRESSIVE VARIANT

- **Suspect**: low/undetectable PSA + rapid visceral, lytic, brain or bulky pelvic disease.
- **Hypercalcaemia via PTHrP**; low PSMA; chromogranin A not required.
- **Molecular**: ≥2 of PTEN loss, TP53 mutation, RB1 loss.

- **Treat**: cisplatin–etoposide or taxane–carboplatin ± RT; biopsy first.
- **Radiotherapy**: brain metastases, bulky or critical-site disease.
- **Survival <12 mo**.

ANDROGEN SIGNALING INHIBITORS

- **Adrenal androgens**: up to 10% of total; untouched by LHRH agents.
- **Only approved CYP17 inhibitors**: abiraterone (and Yonsa); orteronel failed.
- **Abiraterone toxicity**: hypokalaemia, hypertension, oedema, LFTs; CV mortality. Correct K⁺ first.
- **Abiraterone with food**: up to 10× exposure; alternative 250 mg with low-fat meal.
- **COU-AA-301** (post-docetaxel): OS 14.8 vs 10.9 mo, HR 0.65.
- **COU-AA-302** (pre-chemo): rPFS ↑; chemo delayed 26.5 vs 16.8 mo.
- **AFFIRM** (post-docetaxel): OS 18.4 vs 13.6 mo, HR 0.63.
- **PREVAIL** (pre-chemo): death HR 0.71; radiographic progression HR 0.19.
- **SPARTAN**: MFS HR 0.28; OS 73.9 vs 59.9 mo.
- **ARAMIS**: MFS 40.4 vs 18.4 mo; OS HR 0.69.
- **PROSPER**: death risk ↓ 27%.
- **Doses**: enzalutamide 160 mg; apalutamide 240 mg; darolutamide **300 mg twice daily**.
- **Enzalutamide**: seizures ~1%; worse with prednisone.
- **Enzalutamide interactions**: avoid CYP2C8 inhibitors; lowers CYP3A4/2C9/2C19 substrates.
- **Apalutamide**: rash, hypothyroidism. **Apalutamide/darolutamide**: no BBB penetration.
- **Falls/fractures**: caution >75 yr.
- **After docetaxel** → ARPI; **after ARPI** → chemo/radium/sipuleucel-T; **after both** → radioligand or cabazitaxel.

NOVEL AR-TARGETING APPROACHES

- **PROTAC degraders**: ARV-110 (PSA50 in T878A/S, H875Y); ARV-766 (L702H).

CYTOTOXIC CHEMOTHERAPY

- **TAX 327**: OS 18.9 (3-weekly) vs 17.3 (weekly) vs 16.4 mo (mitoxantrone); HR 0.76 — first survival gain.
- **Docetaxel schedules**: 3-weekly → neutropenia; weekly → nausea, nails, lacrimation, diarrhoea.
- **SWOG 9916**: + estramustine → thrombosis, no added benefit.
- **TROPIC**: cabazitaxel vs mitoxantrone OS 15.1 vs 12.7 mo, HR 0.70.

- **TROPIC subgroup:** benefit even if progression <3 mo from docetaxel start → little cross-resistance.
- **Cabazitaxel:** poor P-gp affinity; neutropenia 82%; G-CSF; little neuropathy.
- **PROSELICA:** 20 mg/m² non-inferior to 25.
- **FIRSTANA:** cabazitaxel not better than docetaxel first line.
- **CARD:** OS 13.6 vs 11.0 mo, HR 0.64.
- **Mitoxantrone (1997):** palliation only, no survival gain.
- **Platinum:** neuroendocrine; germline BRCA2 (docetaxel-carboplatin).
- **mCSPC triplets:** ADT + docetaxel + abiraterone (PEACE-1) or darolutamide (ARASENS).
- **Failed with docetaxel:** anti-angiogenics, atrasentan, zibotentan, dasatinib, calcitriol.

RADIOPHARMACEUTICALS

- **Radium-223:** alpha, path <100 µm; 50 kBq/kg 4-weekly × 6.
- **ALSYMPCA:** OS HR 0.70, +3.6 mo; SSE HR 0.66; fewer AEs than placebo.
- **Radium eligibility:** no visceral disease, no bulky nodes.
- **Radium + any ARPI** → add zoledronic acid or denosumab.
- **TheraP:** Lu-PSMA vs cabazitaxel PSA response 65% vs 37%; no OS difference.
- **VISION:** rPFS 8.7 vs 3.4 mo (HR 0.40); OS 15.3 vs 11.3 mo (HR 0.62).
- **Lu-PSMA toxicity:** fatigue, **dry mouth**, nausea, anaemia.
- **Lu-PSMA FDA 2022:** post-ARPI and taxane; PSMAfore tests pre-chemotherapy.

PARP INHIBITORS

- **Somatic DNA-repair mutations:** 20–25% of mCRPC.
- **PROfound** (olaparib, BRCA/ATM): rPFS 7.4 vs 3.6 mo, HR 0.34; FDA May 2020, broad HRR panel.
- **Rationale for ARPI + PARPI:** ADT upregulates PARP-dependent repair.
- **TRITON3** (rucaparib, BRCA): rPFS 11.2 vs 6.4 mo; ATM HR 0.95.
- **PROpel** (olaparib + abiraterone): OS 42.1 vs 34.7 mo — longest first-line median.
- **MAGNITUDE** (niraparib + abiraterone): BRCA rPFS 19.5 vs 10.9 mo; no benefit HRR-negative.
- **TALAPRO-2** (talazoparib + enzalutamide): all-comers rPFS HR 0.63; approved HRR-mutated.
- **GALAHAD** (niraparib): single-arm; BRCA rPFS 8.2 mo.
- **Class toxicity:** anaemia, nausea, fatigue; rarely MDS, PE.
- **ORR** (RECIST CR + PR): commonest basis for FDA solid-tumour approval.

NOVEL MOLECULAR APPROACHES

- **ATM-deficient:** ATR inhibitors (synthetic lethality) rather than PARPI.

IMMUNOTHERAPY

- **IMPACT** (sipuleucel-T): OS 25.8 vs 21.7 mo, HR 0.78; no PFS/PSA change.
- **Sipuleucel-T:** autologous dendritic cells + PAP-GM-CSF; FDA April 2010; 64% placebo crossover.

- **Sipuleucel-T eligibility:** asymptomatic/minimally symptomatic; no visceral mets; no opioids.
- **Ipilimumab phase III ×2:** no OS gain (HR 0.85; HR 1.11).
- **Pembrolizumab:** MSI-H/dMMR or TMB ≥10/Mb; 200 mg 3-weekly; dMMR 2–5%.
- **dMMR testing:** NGS or IHC for MSH2, MSH6, MLH1, PMS2.
- **dMMR found** → germline testing for Lynch syndrome.
- **Ipilimumab irAE:** colitis 15–20% commonest.

EMERGING APPROACHES

- **BAT:** testosterone cypionate 400 mg IM 4-weekly; not if opioid pain or bulky visceral disease.
- **TRANSFORMER:** BAT = enzalutamide for PFS; PFS2 28.2 vs 19.6 mo BAT-first.
- **PSMA BiTEs:** dose-limiting CRS → tocilizumab.
- **PTEN loss ~40%;** IPATential150 benefit only PTEN-deficient.

PALLIATIVE MANAGEMENT

- **Dexamethasone:** 16–100 mg/day; load 10 mg IV (20–40 mg PO), then 4–10 mg 6–12-hourly; taper 2–3 wk.
- **Imaging:** spinal MRI (replaces myelography).
- **Operate if:** progression on RT, unstable fracture, post-RT recurrence; surgery + RT > RT alone (Patchell).
- **Zoledronate:** 4 mg IV 4-weekly, renally adjusted; Ca 1000 mg + vitamin D 800 IU.
- **Denosumab:** 120 mg SC 4-weekly; no renal adjustment; ONJ 2–4%.
- **Denosumab vs zoledronate:** first SRE 20.7 vs 17.1 mo; no OS difference.
- **ONJ risk:** dental work, poor dentition — avoid zoledronate.
- **Hypercalcaemia rare** → think neuroendocrine (PTHrP).
- **Bone agents:** reduce SREs; no antitumour or survival effect.
- **Hydronephrosis:** drainage not needed if contralateral kidney adequate.
- **Circumoral numbness** → skull base → radiotherapy.

CLASSIC TRAPS

- Trap: CRPC is androgen independent — Actually: AR-driven; continue ADT.
- Trap: falling PSA = response — Actually: may be neuroendocrine transformation.
- Trap: new bone-scan lesions at 12 wk = progression — Actually: may be flare; confirm.
- Trap: abiraterone for nmCRPC — Actually: not approved; use apa/enza/daro.
- Trap: radium-223 + abiraterone — Actually: ↑ deaths and fractures.
- Trap: second ARPI after first — Actually: cross-resistance; cabazitaxel wins (CARD).
- Trap: cabazitaxel first line — Actually: not superior to docetaxel (FIRSTANA).
- Trap: mitoxantrone prolongs survival — Actually: palliation only.
- Trap: PARPI for ATM — Actually: weak benefit; rucaparib BRCA only.
- Trap: sipuleucel-T with visceral disease — Actually: not indicated; expect no PSA fall.